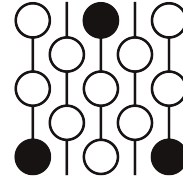


Review of Submission 10C

First Proof Second Batch

Reviewer 1

June 4–8, 2026



1 Review

The solution presented in this article follows a reasonable strategy. However, the proof suffers from various crucial mistakes in the two most important lemmas, rendering the proof to not lift from the ground. I am not sure if the mistakes are easily fixable, but it is plausible that they could be. In any case, it is currently not publishable.

1. The statement and proof of Lemma 3.1 is incorrect, because it is assumed that q_i are central projections, which is not true. Therefore, the fact that $\varphi(\bigwedge_i q_i) = 1$ is totally unjustified. I do not see a way to salvage this as is.
2. Unfortunately, the proof of Lemma 3.2 is also incorrect. In fact, this is the most important lemma for the proof, where there is a non trivial amount of novelty required. Unfortunately it is mentioned that the first assertion is the “support projection form of the graph product bimodule calculation”, which is a totally unjustified and non-rigorous claim. There is no justification for this, and it is by no means a trivial consequence of known results.

Correctness. The proof of the main result is incorrect for the reasons mentioned above, however the strategy is coherent.

Novelty. In the proof of Lemma 3.2 which is the most crucial step in the proof, it is claimed that bimodule calculations can help. While this is not substantiated, it is conceivable that there could be a path towards a proof using this idea. This idea can be considered novel, and suggests an approach which is not in the human solution. However, it is not justified at all, so it cannot be considered as a true novelty.

2 Recommendation

Requires major revisions. At minimum, correct proofs of Lemmas 3.1 and 3.2 are necessary. The rest of the strategy seems to be coherent, which is why I recommend a major revision, as opposed to a straightforward rejection.

3 Submission with inline comments

Abstract

We prove that if Γ is a finite join-irreducible simplicial graph with at least three vertices and if each tracial vertex algebra (M_v, τ_v) contains a trace-zero unitary, then the tracial graph product von Neumann algebra is properly proximal.

3.1 Preliminaries

We use the following standard convention. A finite graph $\Gamma = (\mathcal{V}, \mathcal{E})$ is called *irreducible* if it is not a non-trivial join: equivalently, there is no non-empty proper subset $S \subset \mathcal{V}$ such that every vertex of S is adjacent to every vertex of $\mathcal{V} \setminus S$. Equivalently, the complement graph Γ^c is connected.

For $U \subset \mathcal{V}$, let Γ_U be the induced subgraph and let

$$(M_U, \tau_U) = \bigstar_{v \in \Gamma_U} (M_v, \tau_v)$$

denote the corresponding graph product von Neumann algebra. We put $M_\emptyset = \mathbb{C}$. The canonical trace-preserving conditional expectation is denoted

$$E_U : M \rightarrow M_U.$$

We write

$$\text{Lk}(v) = \{w \in \mathcal{V} : (v, w) \in \mathcal{E}\}, \quad \text{St}(v) = \text{Lk}(v) \cup \{v\}.$$

We use the definition of proper proximality for finite von Neumann algebras introduced by Ding–Kunnawalkam Elayavalli–Peterson [3]. If $N \subset M$ is a von Neumann subalgebra with expectation, let X_N denote the M -boundary piece associated with N , i.e. the boundary piece generated by the Jones projection $e_N : L^2(M) \rightarrow L^2(N)$. The small-at-infinity operator system is denoted $S_{X_N}(M)$. By [3, Theorems 6.2 and 8.5], M is properly proximal relative to X_N exactly when there is no M -central state on $S_{X_N}(M)$ whose restriction to M is normal.

We shall use two standard consequences of the theory in [3].

Lemma 3.1 (Upgrading from relative boundary pieces). *Let (M, τ) be separable and tracial. Let $X_1, \dots, X_n$ be M -boundary pieces and let q_i be the support projection of $K_{X_i}(M)$ in the bidual model of [3, Section 8]. Let $q_{\mathbb{K}}$ be the corresponding support projection for the compact boundary piece $\mathbb{K}(L^2 M)$. Suppose:*

- (i) M is properly proximal relative to every X_i ;
- (ii) $\bigwedge_{i=1}^n q_i = q_{\mathbb{K}}$.

Then M is properly proximal.

Proof. If M were not properly proximal, then [3, Theorem 8.5] would give an M -central state φ on the bidual small-at-infinity system for the compact boundary whose restriction to M is normal. Since proper proximality relative to any X_i rules out amenable direct summands, $\varphi(q_{\mathbb{K}}) = 0$; otherwise Connes' amenability criterion would produce an amenable direct summand.

For each i , if $\varphi(q_i^\perp) > 0$, then the normalized compression of φ by $q_i^\perp$ gives an M -central state on the small-at-infinity system relative to X_i , still normal on M , contradicting relative proper proximality. Hence $\varphi(q_i) = 1$ for all i . Since the q_i are central projections and the family is finite,

$$\varphi\left(\bigwedge_i q_i\right) = 1.$$

By assumption, $\bigwedge_i q_i = q_{\mathbb{K}}$, contradicting $\varphi(q_{\mathbb{K}}) = 0$. $\square$

Lemma 3.2 (Boundary cover for graph products). *For a graph product $M = \star_\Gamma(M_v, \tau_v)$ and subsets $U_1, \dots, U_n \subset \mathcal{V}$, let q_{U_i} be the support projection associated with the boundary piece $X_{M_{U_i}}$. Then*

$$q_{U_1} \wedge \dots \wedge q_{U_n} = q_{U_1 \cap \dots \cap U_n}.$$

In particular,

$$\bigwedge_{v \in \mathcal{V}} q_{\text{Lk}(v)} = q_\emptyset = q_{\mathbb{K}}.$$

Proof. The first assertion is the support-projection form of the graph-product bimodule calculation for subgraph algebras; see Charlesworth–de Santiago–Hayes–Jekel–Kunnawalkam Elayavalli–Nelson [2, Theorem 0.4, Theorem 5.4, and Proposition 5.5]. Equivalently, the M - M boundary piece generated by e_{M_U} intersects/fuses with the one generated by e_{M_W} exactly as the subgraph algebras intersect:

$$M_U \cap M_W = M_{U \cap W}, \quad E_U E_W = E_{U \cap W}.$$

Iterating gives the formula for finitely many U_i .

For $U_i = \text{Lk}(v_i)$ over all vertices, the intersection is empty because $v \notin \text{Lk}(v)$ for every v . Thus the meet is $q_\emptyset$. Since $M_\emptyset = \mathbb{C}$ and the boundary piece generated by $e_{\mathbb{C}}$ is exactly $\mathbb{K}(L^2 M)$, we have $q_\emptyset = q_{\mathbb{K}}$. $\square$

3.2 A relative proper proximality criterion for amalgamated free products

We need the following elementary relative version of the usual paradoxical-decomposition proof for free products.

Proposition 3.3. *Let*

$$(M, \tau) = (A, E_B^A) *_B (C, E_B^C)$$

be a tracial amalgamated free product over a common von Neumann subalgebra B . Suppose there are unitaries

$$u_1, u_2 \in C, \quad w \in A$$

such that

$$E_B^C(u_1) = E_B^C(u_2) = E_B^C(u_1^* u_2) = 0, \quad E_B^A(w) = 0.$$

Then M is properly proximal relative to the boundary piece X_B associated with B .

Proof. Let $A^\circ = \ker E_B^A$ and $C^\circ = \ker E_B^C$. In the standard L^2 -model of the amalgamated free product,

$$L^2(M) = L^2(B) \oplus H_A \oplus H_C,$$

where H_A is the closed span of reduced words beginning in A° , and H_C is the closed span of reduced words beginning in C° . Let P_A and P_C be the corresponding orthogonal projections.

A direct Fock-space computation shows that

$$P_A, P_C \in S_{X_B}(M).$$

Indeed, right multiplication by an algebraic reduced word changes the first letter only through cancellations that factor through $L^2(B)$; hence the commutators with the right action of M lie in the ideal generated by e_B , namely $K_{X_B}(M)$. This is the standard computation used in [3, Section 6].

Assume, toward contradiction, that there exists an M -central state φ on $S_{X_B}(M)$ whose restriction to M is normal. Since $u_1, u_2 \in C^\circ$ and $E_B(u_1^*u_2) = 0$, the subspaces $u_1^*H_A$ and $u_2^*H_A$ are orthogonal and both are contained in H_C . Hence

$$u_1^*P_Au_1 + u_2^*P_Au_2 \leq P_C.$$

By M -centrality and unitarity,

$$2\varphi(P_A) = \varphi(u_1^*P_Au_1) + \varphi(u_2^*P_Au_2) \leq \varphi(P_C).$$

Similarly, since $w \in A^\circ$,

$$w^*P_Cw \leq P_A,$$

and therefore

$$\varphi(P_C) = \varphi(w^*P_Cw) \leq \varphi(P_A).$$

Thus $\varphi(P_A) = \varphi(P_C) = 0$.

Finally, since $u_1 \in C^\circ$, the range of $u_1^*e_Bu_1$ is contained in H_C , so

$$u_1^*e_Bu_1 \leq P_C.$$

Again by centrality,

$$\varphi(e_B) = \varphi(u_1^*e_Bu_1) \leq \varphi(P_C) = 0.$$

But $1 = P_A + P_C + e_B$, so $\varphi(1) = 0$, a contradiction. $\square$

3.3 Graph-product input

For every vertex v , the graph product admits the canonical amalgamated free product decomposition

$$M = M_{\text{St}(v)} *_{M_{\text{Lk}(v)}} M_{\mathcal{V} \setminus \{v\}}.$$

This follows directly from the graph-product normal form and is standard; see [1] and [2, Section 1.1].

We now verify that the hypotheses of Proposition 3.3 hold for each vertex.

Lemma 3.4. *Let Γ be finite, irreducible, and $|\mathcal{V}| \geq 3$. Fix $v \in \mathcal{V}$ and put $B = M_{\text{Lk}(v)}$. In the decomposition*

$$M = M_{\text{St}(v)} *_B M_{\mathcal{V} \setminus \{v\}},$$

there exist a unitary $w \in M_{\text{St}(v)}$ and unitaries $u_1, u_2 \in M_{\mathcal{V} \setminus \{v\}}$ such that

$$E_B(w) = 0, \quad E_B(u_1) = E_B(u_2) = E_B(u_1^* u_2) = 0.$$

Proof. Choose, for each vertex s , a trace-zero unitary $a_s \in M_s$.

First take

$$w = a_v \in M_v \subset M_{\text{St}(v)}.$$

Since $M_{\text{St}(v)} = M_v \overline{\otimes} M_{\text{Lk}(v)}$ over the commuting graph product decomposition, we have

$$E_B(w) = \tau_v(a_v)1 = 0.$$

Let

$$L = \text{Lk}(v), \quad W = \mathcal{V} \setminus (\{v\} \cup L).$$

The set W is non-empty. Indeed, if $W = \emptyset$, then v is adjacent to every vertex in $\mathcal{V} \setminus \{v\}$, giving the non-trivial join decomposition

$$\Gamma = \{v\} + \Gamma_{\mathcal{V} \setminus \{v\}},$$

contrary to irreducibility.

We distinguish two cases.

Case 1: $L = \emptyset$. Then $B = \mathbb{C}$. Since $|\mathcal{V}| \geq 3$, choose distinct vertices $r, s \in \mathcal{V} \setminus \{v\}$. Put

$$u_1 = a_r, \quad u_2 = a_s a_r.$$

Both are unitaries in $M_{\mathcal{V} \setminus \{v\}}$. Their scalar expectations are zero:

$$\tau(u_1) = 0, \quad \tau(u_2) = 0.$$

Moreover,

$$u_1^* u_2 = a_r^* a_s a_r.$$

If r and s are adjacent, this equals a_s , whose trace is zero. If they are not adjacent, it is a non-empty reduced word with trace-zero letters, hence has trace zero. Thus $E_B(u_1^* u_2) = 0$.

Case 2: $L \neq \emptyset$. We claim that there exist $\ell \in L$ and $r \in W$ such that ℓ is not adjacent to r . If not, then every vertex of L is adjacent to every vertex of $\mathcal{V} \setminus L$ because vertices of L are adjacent to v by definition and, by assumption, to every vertex of W . Hence

$$\Gamma = \Gamma_L + \Gamma_{\mathcal{V} \setminus L},$$

again contradicting irreducibility.

Choose such $\ell \in L$ and $r \in W$. Put

$$u_1 = a_r, \quad u_2 = a_\ell a_r.$$

Since $a_\ell \in M_\ell \subset B$ and $r \notin L$, both u_1 and u_2 lie in $M_{\mathcal{V} \setminus \{v\}}$ and satisfy

$$E_B(u_1) = E_B(u_2) = 0.$$

Furthermore,

$$u_1^* u_2 = a_r^* a_\ell a_r.$$

Because r and ℓ are not adjacent, this is a reduced word involving the vertex $r \notin L$, so its expectation onto $B = M_L$ is zero. Hence $E_B(u_1^* u_2) = 0$. $\square$

Corollary 3.5. *For every $v \in \mathcal{V}$, the graph product M is properly proximal relative to the boundary piece $X_{M_{\text{Lk}(v)}}$.*

Proof. Apply Proposition 3.3 to

$$M = M_{\text{St}(v)} \underset{M_{\text{Lk}(v)}}{*} M_{\mathcal{V} \setminus \{v\}},$$

using the unitaries supplied by Lemma 3.4. $\square$

3.4 Proof of the theorem

Theorem 3.6. *Let Γ be a finite simplicial graph with $|\mathcal{V}| \geq 3$. Let (M_v, τ_v) be a separable tracial von Neumann algebra for every $v \in \mathcal{V}$. Assume:*

- (a) Γ is irreducible, i.e. not a non-trivial join;
- (b) every M_v contains a trace-zero unitary.

Let

$$(M, \tau) = \bigstar_{v \in \Gamma} (M_v, \tau_v)$$

be the tracial graph product von Neumann algebra. Then M is properly proximal.

Proof. For every vertex v , Corollary 3.5 shows that M is properly proximal relative to the boundary piece $X_{M_{\text{Lk}(v)}}$.

Therefore M has no amenable direct summand: if pM were an amenable direct summand, Connes' amenability criterion would give an M -central state on $\mathbb{B}(L^2(pM))$, hence by extension an M -central state on every $S_{X_{M_{\text{Lk}(v)}}}(M)$ normal on M , contradicting relative proper proximality.

By Lemma 3.2,

$$\bigwedge_{v \in \mathcal{V}} q_{\text{Lk}(v)} = q_{\mathbb{K}}.$$

Thus the finite family of boundary pieces $\{X_{M_{\text{Lk}(v)}} : v \in \mathcal{V}\}$ satisfies the hypotheses of the upgrading Lemma 3.1. Consequently M is properly proximal. $\square$

References

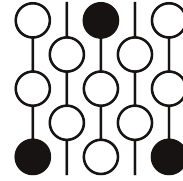
- [1] M. Caspers and P. Fima, *Graph products of operator algebras*, J. Noncommut. Geom. **11** (2017), no. 1, 367–411.
- [2] I. Charlesworth, R. de Santiago, B. Hayes, D. Jekel, S. Kunnawalkam Elayavalli and B. Nelson, *On the structure of graph product von Neumann algebras*, Publ. Res. Inst. Math. Sci. **61** (2025), no. 4, 713–762.
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Review of Submission 10C

First Proof Second Batch

Reviewer 2

June 4–8, 2026



1 Review

Correctness. The solution gives a correct outline but has one lemma which is wrong as stated (but can be reasonably easily fixed) and another lemma which is correct but effectively not proved. The issue with the second lemma should be considered a major gap which is highly nontrivial to fix.

Novelty. The solution is novel in that it suggests an approach that is different from the human solution. But the solution does not prove the main technical result needed. It can be made to work with significant additional efforts, but this is not obvious at all and no justification exists within the solution itself.

2 Recommendation

This is between requiring major revisions and outright rejection. The solution does not prove at all the main technical result needed, instead citing a paper and supplying no hints as to how to apply the result there to obtain the needed technical lemma. It turns out that this approach works, but this is by no means obvious or routine and requires a significant amount of additional arguments.

3 Submission with inline comments

Abstract

We prove that if Γ is a finite join-irreducible simplicial graph with at least three vertices and if each tracial vertex algebra (M_v, τ_v) contains a trace-zero unitary, then the tracial graph product von Neumann algebra is properly proximal.

3.1 Preliminaries

We use the following standard convention. A finite graph $\Gamma = (\mathcal{V}, \mathcal{E})$ is called *irreducible* if it is not a non-trivial join: equivalently, there is no non-empty proper subset $S \subset \mathcal{V}$ such that every vertex of S is adjacent to every vertex of $\mathcal{V} \setminus S$. Equivalently, the complement graph Γ^c is connected.

For $U \subset \mathcal{V}$, let Γ_U be the induced subgraph and let

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denote the corresponding graph product von Neumann algebra. We put $M_\emptyset = \mathbb{C}$. The canonical trace-preserving conditional expectation is denoted

$$E_U : M \rightarrow M_U.$$

We write

$$\text{Lk}(v) = \{w \in \mathcal{V} : (v, w) \in \mathcal{E}\}, \quad \text{St}(v) = \text{Lk}(v) \cup \{v\}.$$

We use the definition of proper proximality for finite von Neumann algebras introduced by Ding–Kunnawalkam Elayavalli–Peterson [3]. If $N \subset M$ is a von Neumann subalgebra with expectation, let X_N denote the M -boundary piece associated with N , i.e. the boundary piece generated by the Jones projection $e_N : L^2(M) \rightarrow L^2(N)$. The small-at-infinity operator system is denoted $S_{X_N}(M)$. By [3, Theorems 6.2 and 8.5], M is properly proximal relative to X_N exactly when there is no M -central state on $S_{X_N}(M)$ whose restriction to M is normal.

We shall use two standard consequences of the theory in [3].

Lemma 3.1 (Upgrading from relative boundary pieces). *Let (M, τ) be separable and tracial. Let $X_1, \dots, X_n$ be M -boundary pieces and let q_i be the support projection of $K_{X_i}(M)$ in the bidual model of [3, Section 8]. Let $q_{\mathbb{K}}$ be the corresponding support projection for the compact boundary piece $\mathbb{K}(L^2 M)$. Suppose:*

- (i) M is properly proximal relative to every X_i ;
- (ii) $\bigwedge_{i=1}^n q_i = q_{\mathbb{K}}$.

Then M is properly proximal.

Proof. If M were not properly proximal, then [3, Theorem 8.5] would give an M -central state φ on the bidual small-at-infinity system for the compact boundary whose restriction to M is normal. Since proper proximality relative to any X_i rules out amenable direct summands, $\varphi(q_{\mathbb{K}}) = 0$; otherwise Connes' amenability criterion would produce an amenable direct summand.

Reviewer 2

While the result of this part is correct, it requires more justification. In particular, while $q_{\mathbb{K}}$ is in the small-at-infinity boundary. The same issue appears in the first line of the next paragraph, namely, justification is needed to show q_i is in the small-at-infinity boundary (for the compact boundary), as otherwise $\varphi(q_i^\perp)$ makes no sense.

For each i , if $\varphi(q_i^\perp) > 0$, then the normalized compression of φ by $q_i^\perp$ gives an M -central state on the small-at-infinity system relative to X_i , still normal on M , contradicting relative proper proximality. Hence $\varphi(q_i) = 1$ for all i . Since the q_i are central projections and the family is finite,

$$\varphi\left(\bigwedge_i q_i\right) = 1.$$

This part is wrong. the claim that q_i are central projections are wrong, so the argument does not work. In fact, the lemma is wrong as stated. It needs the additional assumption that q_i commute with each other, which is also the ingredient needed to make this part of the argument work.

By assumption, $\bigwedge_i q_i = q_{\mathbb{K}}$, contradicting $\varphi(q_{\mathbb{K}}) = 0$. $\square$

Lemma 3.2 (Boundary cover for graph products). *For a graph product $M = \star_{\Gamma}(M_v, \tau_v)$ and subsets $U_1, \dots, U_n \subset \mathcal{V}$, let q_{U_i} be the support projection associated with the boundary piece $X_{M_{U_i}}$. Then*

$$q_{U_1} \wedge \dots \wedge q_{U_n} = q_{U_1 \cap \dots \cap U_n}.$$

In particular,

$$\bigwedge_{v \in \mathcal{V}} q_{\text{Lk}(v)} = q_{\emptyset} = q_{\mathbb{K}}.$$

Proof. The first assertion is the support-projection form of the graph-product bimodule calculation for subgraph algebras; see Charlesworth–de Santiago–Hayes–Jekel–Kunnawalkam Elayavalli–Nelson [2, Theorem 0.4, Theorem 5.4, and Proposition 5.5]. Equivalently, the M - M boundary piece generated by e_{M_U} intersects/fuses with the one generated by e_{M_W} exactly as the subgraph algebras intersect:

$$M_U \cap M_W = M_{U \cap W}, \quad E_U E_W = E_{U \cap W}.$$

Iterating gives the formula for finitely many U_i .

For $U_i = \text{Lk}(v_i)$ over all vertices, the intersection is empty because $v \notin \text{Lk}(v)$ for every v . Thus the meet is $q_{\emptyset}$. Since $M_{\emptyset} = \mathbb{C}$ and the boundary piece generated by $e_{\mathbb{C}}$ is exactly $\mathbb{K}(L^2 M)$, we have $q_{\emptyset} = q_{\mathbb{K}}$. $\square$

While the AI suggested an approach different from the human solution here (and it turns out that the approach works, per the solution given by another AI system), it gives no justification as to why the “graph-product bimodule calculation” should yield the claimed result, nor is it obvious. The argument for this needs at least around three pages and requires plenty of nontrivial arguments. It should also be noted that, while the lemma is correct as stated, because of the issue in Lemma 3.1, we actually need the additional result that q_{U_i} commutes with each other, a strengthening of the lemma claimed here.

3.2 A relative proper proximality criterion for amalgamated free products

We need the following elementary relative version of the usual paradoxical-decomposition proof for free products.

Proposition 3.3. *Let*

$$(M, \tau) = (A, E_B^A) *_B (C, E_B^C)$$

be a tracial amalgamated free product over a common von Neumann subalgebra B . Suppose there are unitaries

$$u_1, u_2 \in C, \quad w \in A$$

such that

$$E_B^C(u_1) = E_B^C(u_2) = E_B^C(u_1^*u_2) = 0, \quad E_B^A(w) = 0.$$

Then M is properly proximal relative to the boundary piece X_B associated with B .

Proof. Let $A^\circ = \ker E_B^A$ and $C^\circ = \ker E_B^C$. In the standard L^2 -model of the amalgamated free product,

$$L^2(M) = L^2(B) \oplus H_A \oplus H_C,$$

where H_A is the closed span of reduced words beginning in A° , and H_C is the closed span of reduced words beginning in C° . Let P_A and P_C be the corresponding orthogonal projections.

A direct Fock-space computation shows that

$$P_A, P_C \in S_{X_B}(M).$$

Indeed, right multiplication by an algebraic reduced word changes the first letter only through cancellations that factor through $L^2(B)$; hence the commutators with the right action of M lie in the ideal generated by e_B , namely $K_{X_B}(M)$. This is the standard computation used in [3, Section 6].

Assume, toward contradiction, that there exists an M -central state φ on $S_{X_B}(M)$ whose restriction to M is normal. Since $u_1, u_2 \in C^\circ$ and $E_B(u_1^*u_2) = 0$, the subspaces $u_1^*H_A$ and $u_2^*H_A$ are orthogonal and both are contained in H_C . Hence

$$u_1^*P_Au_1 + u_2^*P_Au_2 \leq P_C.$$

By M -centrality and unitarity,

$$2\varphi(P_A) = \varphi(u_1^*P_Au_1) + \varphi(u_2^*P_Au_2) \leq \varphi(P_C).$$

Similarly, since $w \in A^\circ$,

$$w^*P_Cw \leq P_A,$$

and therefore

$$\varphi(P_C) = \varphi(w^*P_Cw) \leq \varphi(P_A).$$

Thus $\varphi(P_A) = \varphi(P_C) = 0$.

Finally, since $u_1 \in C^\circ$, the range of $u_1^*e_Bu_1$ is contained in H_C , so

$$u_1^*e_Bu_1 \leq P_C.$$

Again by centrality,

$$\varphi(e_B) = \varphi(u_1^*e_Bu_1) \leq \varphi(P_C) = 0.$$

But $1 = P_A + P_C + e_B$, so $\varphi(1) = 0$, a contradiction. □

3.3 Graph-product input

For every vertex v , the graph product admits the canonical amalgamated free product decomposition

$$M = M_{\text{St}(v)} {}_{M_{\text{Lk}(v)}}^* M_{\mathcal{V} \setminus \{v\}}.$$

This follows directly from the graph-product normal form and is standard; see [1] and [2, Section 1.1].

We now verify that the hypotheses of Proposition 3.3 hold for each vertex.

Lemma 3.4. *Let Γ be finite, irreducible, and $|\mathcal{V}| \geq 3$. Fix $v \in \mathcal{V}$ and put $B = M_{\text{Lk}(v)}$. In the decomposition*

$$M = M_{\text{St}(v)} {}_B^* M_{\mathcal{V} \setminus \{v\}},$$

there exist a unitary $w \in M_{\text{St}(v)}$ and unitaries $u_1, u_2 \in M_{\mathcal{V} \setminus \{v\}}$ such that

$$E_B(w) = 0, \quad E_B(u_1) = E_B(u_2) = E_B(u_1^* u_2) = 0.$$

Proof. Choose, for each vertex s , a trace-zero unitary $a_s \in M_s$.

First take

$$w = a_v \in M_v \subset M_{\text{St}(v)}.$$

Since $M_{\text{St}(v)} = M_v \overline{\otimes} M_{\text{Lk}(v)}$ over the commuting graph product decomposition, we have

$$E_B(w) = \tau_v(a_v)1 = 0.$$

Let

$$L = \text{Lk}(v), \quad W = \mathcal{V} \setminus (\{v\} \cup L).$$

The set W is non-empty. Indeed, if $W = \emptyset$, then v is adjacent to every vertex in $\mathcal{V} \setminus \{v\}$, giving the non-trivial join decomposition

$$\Gamma = \{v\} + \Gamma_{\mathcal{V} \setminus \{v\}},$$

contrary to irreducibility.

We distinguish two cases.

Case 1: $L = \emptyset$. Then $B = \mathbb{C}$. Since $|\mathcal{V}| \geq 3$, choose distinct vertices $r, s \in \mathcal{V} \setminus \{v\}$. Put

$$u_1 = a_r, \quad u_2 = a_s a_r.$$

Both are unitaries in $M_{\mathcal{V} \setminus \{v\}}$. Their scalar expectations are zero:

$$\tau(u_1) = 0, \quad \tau(u_2) = 0.$$

Moreover,

$$u_1^* u_2 = a_r^* a_s a_r.$$

If r and s are adjacent, this equals a_s , whose trace is zero. If they are not adjacent, it is a non-empty reduced word with trace-zero letters, hence has trace zero. Thus $E_B(u_1^* u_2) = 0$.

Case 2: $L \neq \emptyset$. We claim that there exist $\ell \in L$ and $r \in W$ such that ℓ is not adjacent to r . If not, then every vertex of L is adjacent to every vertex of $\mathcal{V} \setminus L$ because vertices of L are adjacent to v by definition and, by assumption, to every vertex of W . Hence

$$\Gamma = \Gamma_L + \Gamma_{\mathcal{V} \setminus L},$$

again contradicting irreducibility.

Choose such $\ell \in L$ and $r \in W$. Put

$$u_1 = a_r, \quad u_2 = a_\ell a_r.$$

Since $a_\ell \in M_\ell \subset B$ and $r \notin L$, both u_1 and u_2 lie in $M_{\mathcal{V} \setminus \{v\}}$ and satisfy

$$E_B(u_1) = E_B(u_2) = 0.$$

Furthermore,

$$u_1^* u_2 = a_r^* a_\ell a_r.$$

Because r and ℓ are not adjacent, this is a reduced word involving the vertex $r \notin L$, so its expectation onto $B = M_L$ is zero. Hence $E_B(u_1^* u_2) = 0$. $\square$

Corollary 3.5. *For every $v \in \mathcal{V}$, the graph product M is properly proximal relative to the boundary piece $X_{M_{\text{Lk}(v)}}$.*

Proof. Apply Proposition 3.3 to

$$M = M_{\text{St}(v)} M_{M_{\text{Lk}(v)}}^* M_{\mathcal{V} \setminus \{v\}},$$

using the unitaries supplied by Lemma 3.4. $\square$

3.4 Proof of the theorem

Theorem 3.6. *Let Γ be a finite simplicial graph with $|\mathcal{V}| \geq 3$. Let (M_v, τ_v) be a separable tracial von Neumann algebra for every $v \in \mathcal{V}$. Assume:*

- (a) Γ is irreducible, i.e. not a non-trivial join;
- (b) every M_v contains a trace-zero unitary.

Let

$$(M, \tau) = \bigstar_{v \in \Gamma} (M_v, \tau_v)$$

be the tracial graph product von Neumann algebra. Then M is properly proximal.

Proof. For every vertex v , Corollary 3.5 shows that M is properly proximal relative to the boundary piece $X_{M_{\text{Lk}(v)}}$.

Therefore M has no amenable direct summand: if pM were an amenable direct summand, Connes' amenability criterion would give an M -central state on $\mathbb{B}(L^2(pM))$, hence by extension an M -central state on every $S_{X_{M_{\text{Lk}(v)}}}(M)$ normal on M , contradicting relative proper proximality.

By Lemma 3.2,

$$\bigwedge_{v \in \mathcal{V}} q_{\text{Lk}(v)} = q_{\mathbb{K}}.$$

Thus the finite family of boundary pieces $\{X_{M_{\text{Lk}(v)}} : v \in \mathcal{V}\}$ satisfies the hypotheses of the upgrading Lemma 3.1. Consequently M is properly proximal. $\square$

References

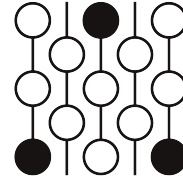
- [1] M. Caspers and P. Fima, *Graph products of operator algebras*, J. Noncommut. Geom. **11** (2017), no. 1, 367–411.
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Review of Submission 10C

First Proof Second Batch

Reviewer 3

June 4–8, 2026



1 Review

This generated proof provides a plausible strategy that is close to the human generated solution. However, several key arguments are not justified and there are gaps inside the proofs. The major gap is in Lemma 3.2, where it wrongly claims that the bimodule calculations in [2, Theorem 0.4, Theorem 5.4, and Proposition 5.5] imply that q_i commutes while the cited contents are not clearly related.

Another major gap is about the usage of bidual. The definition of the bidual and bidual small-at-infinity boundary is never defined. In the preliminaries, the small-at-infinity boundary $S_{X_N}(M)$ is not defined, it only says 'the small-at-infinity operator system', which perhaps means the usual small-at-infinity boundary in $B(L^2M)$. However, in later proofs, the interpretation of $S_{X_N}(M)$ is not fixed: sometimes it implicitly means the bidual version of the small-at-infinity boundary, while sometimes it means the usual small-at-infinity boundary.

Correctness. The overall strategy to show properly proximal is correct, but many statements in the proofs are not justified,

For Lemma 3.1, the proof of the key step $\varphi(\wedge_i q_i) = 1$ is wrong as q_i are not central projections.

For Lemma 3.2, the proof claims that the first assertion follows from bimodule calculation. But this is not true, or at least not obvious at all. In fact, this is the main difficulty of the problem, but the proof skipped it without a good justification.

Novelty. The presented arguments are standard and are close to human proof. The proof of Lemma 3.4 has a degree of novelty, but it makes the proof longer.

2 Recommendation

Requires major revisions on Lemma 3.1 and Lemma 3.2 as the proofs are incorrect.

3 Submission with inline comments

Abstract

We prove that if Γ is a finite join-irreducible simplicial graph with at least three vertices and if each tracial vertex algebra (M_v, τ_v) contains a trace-zero unitary, then the tracial graph product von Neumann algebra is properly proximal.

3.1 Preliminaries

We use the following standard convention. A finite graph $\Gamma = (\mathcal{V}, \mathcal{E})$ is called *irreducible* if it is not a non-trivial join: equivalently, there is no non-empty proper subset $S \subset \mathcal{V}$ such that every vertex of S is adjacent to every vertex of $\mathcal{V} \setminus S$. Equivalently, the complement graph Γ^c is connected.

For $U \subset \mathcal{V}$, let Γ_U be the induced subgraph and let

$$(M_U, \tau_U) = \bigstar_{v \in \Gamma_U} (M_v, \tau_v)$$

denote the corresponding graph product von Neumann algebra. We put $M_\emptyset = \mathbb{C}$. The canonical trace-preserving conditional expectation is denoted

$$E_U : M \rightarrow M_U.$$

We write

$$\text{Lk}(v) = \{w \in \mathcal{V} : (v, w) \in \mathcal{E}\}, \quad \text{St}(v) = \text{Lk}(v) \cup \{v\}.$$

We use the definition of proper proximality for finite von Neumann algebras introduced by Ding–Kunnawalkam Elayavalli–Peterson [3]. If $N \subset M$ is a von Neumann subalgebra with expectation, let X_N denote the M -boundary piece associated with N , i.e. the boundary piece generated by the Jones projection $e_N : L^2(M) \rightarrow L^2(N)$. The small-at-infinity operator system is denoted $S_{X_N}(M)$. By [3, Theorems 6.2 and 8.5], M is properly proximal relative to X_N exactly when there is no M -central state on $S_{X_N}(M)$ whose restriction to M is normal.

We shall use two standard consequences of the theory in [3].

Lemma 3.1 (Upgrading from relative boundary pieces). *Let (M, τ) be separable and tracial. Let $X_1, \dots, X_n$ be M -boundary pieces and let q_i be the support projection of $K_{X_i}(M)$ in the bidual model of [3, Section 8]. Let $q_{\mathbb{K}}$ be the corresponding support projection for the compact boundary piece $\mathbb{K}(L^2 M)$. Suppose:*

- (i) M is properly proximal relative to every X_i ;
- (ii) $\bigwedge_{i=1}^n q_i = q_{\mathbb{K}}$.

Then M is properly proximal.

Proof. If M were not properly proximal, then [3, Theorem 8.5] would give an M -central state φ on the bidual small-at-infinity system for the compact boundary whose restriction to M is normal. Since proper proximality relative to any X_i rules out amenable direct summands, $\varphi(q_{\mathbb{K}}) = 0$; otherwise Connes' amenability criterion would produce an amenable direct summand.

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It is not clear why 'Connes' amenability criterion would produce an amenable direct summand'. 'Connes' amenability criterion' is too vague as there are many amenability criterions due to Connes.

For each i , if $\varphi(q_i^\perp) > 0$, then the normalized compression of φ by $q_i^\perp$ gives an M -central state on the small-at-infinity system relative to X_i , still normal on M , contradicting relative proper proximality. Hence $\varphi(q_i) = 1$ for all i . Since the q_i are central projections and the family is finite,

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q_i are never central projections.

$$\varphi\left(\bigwedge_i q_i\right) = 1.$$

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It is not known whether φ is normal on the whole small-at-infinity boundary and therefore this claim is not justified.

By assumption, $\bigwedge_i q_i = q_{\mathbb{K}}$, contradicting $\varphi(q_{\mathbb{K}}) = 0$. $\square$

Lemma 3.2 (Boundary cover for graph products). *For a graph product $M = \star_\Gamma(M_v, \tau_v)$ and subsets $U_1, \dots, U_n \subset \mathcal{V}$, let q_{U_i} be the support projection associated with the boundary piece $X_{M_{U_i}}$. Then*

$$q_{U_1} \wedge \dots \wedge q_{U_n} = q_{U_1 \cap \dots \cap U_n}.$$

In particular,

$$\bigwedge_{v \in \mathcal{V}} q_{\text{Lk}(v)} = q_\emptyset = q_{\mathbb{K}}.$$

Proof. The first assertion is the support-projection form of the graph-product bimodule calculation for subgraph algebras; see Charlesworth–de Santiago–Hayes–Jekel–Kunnawalkam Elayavalli–Nelson [2, Theorem 0.4, Theorem 5.4, and Proposition 5.5]. Equivalently, the M - M boundary piece generated by e_{M_U} intersects/fuses with the one generated by e_{M_W} exactly as the subgraph algebras intersect:

$$M_U \cap M_W = M_{U \cap W}, \quad E_U E_W = E_{U \cap W}.$$

Iterating gives the formula for finitely many U_i .

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This statement is trying to get around the technical proof by citing the reference. However, the cited paper does not contain anything about the computation of M - M boundary piece. The relation between bimodule and M - M boundary piece is also unclear.

For $U_i = \text{Lk}(v_i)$ over all vertices, the intersection is empty because $v \notin \text{Lk}(v)$ for every v . Thus the meet is $q_\emptyset$. Since $M_\emptyset = \mathbb{C}$ and the boundary piece generated by $e_{\mathbb{C}}$ is exactly $\mathbb{K}(L^2 M)$, we have $q_\emptyset = q_{\mathbb{K}}$. $\square$

3.2 A relative proper proximality criterion for amalgamated free products

We need the following elementary relative version of the usual paradoxical-decomposition proof for free products.

Proposition 3.3. *Let*

$$(M, \tau) = (A, E_B^A) *_B (C, E_B^C)$$

be a tracial amalgamated free product over a common von Neumann subalgebra B . Suppose there are unitaries

$$u_1, u_2 \in C, \quad w \in A$$

such that

$$E_B^C(u_1) = E_B^C(u_2) = E_B^C(u_1^*u_2) = 0, \quad E_B^A(w) = 0.$$

Then M is properly proximal relative to the boundary piece X_B associated with B .

Proof. Let $A^\circ = \ker E_B^A$ and $C^\circ = \ker E_B^C$. In the standard L^2 -model of the amalgamated free product,

$$L^2(M) = L^2(B) \oplus H_A \oplus H_C,$$

where H_A is the closed span of reduced words beginning in A° , and H_C is the closed span of reduced words beginning in C° . Let P_A and P_C be the corresponding orthogonal projections.

A direct Fock-space computation shows that

$$P_A, P_C \in S_{X_B}(M).$$

Indeed, right multiplication by an algebraic reduced word changes the first letter only through cancellations that factor through $L^2(B)$; hence the commutators with the right action of M lie in the ideal generated by e_B , namely $K_{X_B}(M)$. This is the standard computation used in [3, Section 6].

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This paragraph, while strategically correct, need more details about the computation to show that $P_A, P_C \in S_{X_B}(M)$.

Assume, toward contradiction, that there exists an M -central state φ on $S_{X_B}(M)$ whose restriction to M is normal. Since $u_1, u_2 \in C^\circ$ and $E_B(u_1^*u_2) = 0$, the subspaces $u_1^*H_A$ and $u_2^*H_A$ are orthogonal and both are contained in H_C . Hence

$$u_1^*P_Au_1 + u_2^*P_Au_2 \leq P_C.$$

By M -centrality and unitarity,

$$2\varphi(P_A) = \varphi(u_1^*P_Au_1) + \varphi(u_2^*P_Au_2) \leq \varphi(P_C).$$

Similarly, since $w \in A^\circ$,

$$w^*P_Cw \leq P_A,$$

and therefore

$$\varphi(P_C) = \varphi(w^* P_C w) \leq \varphi(P_A).$$

Thus $\varphi(P_A) = \varphi(P_C) = 0$.

Finally, since $u_1 \in C^\circ$, the range of $u_1^* e_B u_1$ is contained in H_C , so

$$u_1^* e_B u_1 \leq P_C.$$

Again by centrality,

$$\varphi(e_B) = \varphi(u_1^* e_B u_1) \leq \varphi(P_C) = 0.$$

But $1 = P_A + P_C + e_B$, so $\varphi(1) = 0$, a contradiction. $\square$

3.3 Graph-product input

For every vertex v , the graph product admits the canonical amalgamated free product decomposition

$$M = M_{\text{St}(v)} *_{M_{\text{Lk}(v)}} M_{\mathcal{V} \setminus \{v\}}.$$

This follows directly from the graph-product normal form and is standard; see [1] and [2, Section 1.1].

We now verify that the hypotheses of Proposition 3.3 hold for each vertex.

Lemma 3.4. *Let Γ be finite, irreducible, and $|\mathcal{V}| \geq 3$. Fix $v \in \mathcal{V}$ and put $B = M_{\text{Lk}(v)}$. In the decomposition*

$$M = M_{\text{St}(v)} *_B M_{\mathcal{V} \setminus \{v\}},$$

there exist a unitary $w \in M_{\text{St}(v)}$ and unitaries $u_1, u_2 \in M_{\mathcal{V} \setminus \{v\}}$ such that

$$E_B(w) = 0, \quad E_B(u_1) = E_B(u_2) = E_B(u_1^* u_2) = 0.$$

Proof. Choose, for each vertex s , a trace-zero unitary $a_s \in M_s$.

First take

$$w = a_v \in M_v \subset M_{\text{St}(v)}.$$

Since $M_{\text{St}(v)} = M_v \overline{\otimes} M_{\text{Lk}(v)}$ over the commuting graph product decomposition, we have

$$E_B(w) = \tau_v(a_v)1 = 0.$$

Let

$$L = \text{Lk}(v), \quad W = \mathcal{V} \setminus (\{v\} \cup L).$$

The set W is non-empty. Indeed, if $W = \emptyset$, then v is adjacent to every vertex in $\mathcal{V} \setminus \{v\}$, giving the non-trivial join decomposition

$$\Gamma = \{v\} + \Gamma_{\mathcal{V} \setminus \{v\}},$$

contrary to irreducibility.

We distinguish two cases.

Case 1: $L = \emptyset$. Then $B = \mathbb{C}$. Since $|\mathcal{V}| \geq 3$, choose distinct vertices $r, s \in \mathcal{V} \setminus \{v\}$. Put

$$u_1 = a_r, \quad u_2 = a_s a_r.$$

Both are unitaries in $M_{\mathcal{V} \setminus \{v\}}$. Their scalar expectations are zero:

$$\tau(u_1) = 0, \quad \tau(u_2) = 0.$$

Moreover,

$$u_1^* u_2 = a_r^* a_s a_r.$$

If r and s are adjacent, this equals a_s , whose trace is zero. If they are not adjacent, it is a non-empty reduced word with trace-zero letters, hence has trace zero. Thus $E_B(u_1^* u_2) = 0$.

Case 2: $L \neq \emptyset$. We claim that there exist $\ell \in L$ and $r \in W$ such that ℓ is not adjacent to r . If not, then every vertex of L is adjacent to every vertex of $\mathcal{V} \setminus L$ because vertices of L are adjacent to v by definition and, by assumption, to every vertex of W . Hence

$$\Gamma = \Gamma_L + \Gamma_{\mathcal{V} \setminus L},$$

again contradicting irreducibility.

Choose such $\ell \in L$ and $r \in W$. Put

$$u_1 = a_r, \quad u_2 = a_\ell a_r.$$

Since $a_\ell \in M_\ell \subset B$ and $r \notin L$, both u_1 and u_2 lie in $M_{\mathcal{V} \setminus \{v\}}$ and satisfy

$$E_B(u_1) = E_B(u_2) = 0.$$

Furthermore,

$$u_1^* u_2 = a_r^* a_\ell a_r.$$

Because r and ℓ are not adjacent, this is a reduced word involving the vertex $r \notin L$, so its expectation onto $B = M_L$ is zero. Hence $E_B(u_1^* u_2) = 0$. $\square$

Corollary 3.5. *For every $v \in \mathcal{V}$, the graph product M is properly proximal relative to the boundary piece $X_{M_{\text{Lk}(v)}}$.*

Proof. Apply Proposition 3.3 to

$$M = M_{\text{St}(v)} {}_{M_{\text{Lk}(v)}}^* M_{\mathcal{V} \setminus \{v\}},$$

using the unitaries supplied by Lemma 3.4. $\square$

3.4 Proof of the theorem

Theorem 3.6. *Let Γ be a finite simplicial graph with $|\mathcal{V}| \geq 3$. Let (M_v, τ_v) be a separable tracial von Neumann algebra for every $v \in \mathcal{V}$. Assume:*

- (a) Γ is irreducible, i.e. not a non-trivial join;
- (b) every M_v contains a trace-zero unitary.

Let

$$(M, \tau) = \bigstar_{v \in \Gamma} (M_v, \tau_v)$$

be the tracial graph product von Neumann algebra. Then M is properly proximal.

Proof. For every vertex v , Corollary 3.5 shows that M is properly proximal relative to the boundary piece $X_{M_{\text{Lk}(v)}}$.

Therefore M has no amenable direct summand: if pM were an amenable direct summand, Connes' amenability criterion would give an M -central state on $\mathbb{B}(L^2(pM))$, hence by extension an M -central state on every $S_{X_{M_{\text{Lk}(v)}}}(M)$ normal on M , contradicting relative proper proximality.

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The claim that the M -central state extends to $S_{X_{M_{\text{Lk}(v)}}}(M)$ is not proven.

By Lemma 3.2,

$$\bigwedge_{v \in \mathcal{V}} q_{\text{Lk}(v)} = q_{\mathbb{K}}.$$

Thus the finite family of boundary pieces $\{X_{M_{\text{Lk}(v)}} : v \in \mathcal{V}\}$ satisfies the hypotheses of the upgrading Lemma 3.1. Consequently M is properly proximal. $\square$

References

- [1] M. Caspers and P. Fima, *Graph products of operator algebras*, J. Noncommut. Geom. **11** (2017), no. 1, 367–411.
- [2] I. Charlesworth, R. de Santiago, B. Hayes, D. Jekel, S. Kunnawalkam Elayavalli and B. Nelson, *On the structure of graph product von Neumann algebras*, Publ. Res. Inst. Math. Sci. **61** (2025), no. 4, 713–762.
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